



c23-c-106

23037

BOARD DIPLOMA EXAMINATION, (C-23)

OCTOBER/NOVEMBER—2024

DCE – FIRST YEAR EXAMINATION

SURVEYING—I

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answer should be brief and straight to the point and shall not exceed five simple sentences.

1. Define surveying and state the three purposes of conducting surveying.
2. Define the following terms :
 - (a) Base line
 - (b) Check line
 - (c) Tie line
3. State the principle of chain surveying.
4. Define local attraction. How do you detect it?
5. Convert WCB into quadrantal bearing for
 - (a) $75^{\circ}45'$
 - (b) $130^{\circ}15'$
6. Define the following terms :
 - (a) Back sight
 - (b) Fore sight
7. List the fundamental lines of a dumpy level.

8. Define the following terms :
 - (a) Contour interval
 - (b) Horizontal equivalent
9. List any three types of benchmarks.
10. State any three uses of Abney level.

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answers.

11. Explain the classification of surveys based on instruments used in detail.
12. Define ranging and explain the method of reciprocal ranging with a sketch.
13. The following are the perpendicular offsets taken at 15 m intervals from a survey line *AB* to an irregular boundary line :
 3·10 m, 4·20 m, 5·35 m, 6·45 m, 7·15 m, 8·25 m, 7·95 m, 5·20 m and 5·50 m
 Calculate the area enclosed between the survey line, the irregular boundary, the first and last offsets by (i) Trapezoidal rule and (ii) Simpson's rule.
14. Explain different types of errors in a compass surveying.
15. The bearings observed in a closed traverse survey at a place where local attraction was suspected are given below :

Line	Fore bearing	Back bearing
<i>AB</i>	124°30'	304°30'
<i>BC</i>	68°15'	246°00'
<i>CD</i>	310°30'	135°15'
<i>DA</i>	200°15'	17°45'

At what stations do you suspect local attraction? Find the corrected bearing.

- 16.** The following consecutive readings were taken with a dumpy level :
1·585 m; 1·925 m; 2·350 m; 0·220 m; 2·655 m; 2·420 m; 2·925 m; 2·580 m;
1·235 m; 2·995 m. The instrument was shifted after the fourth and the
sixth readings. The first reading was taken on the staff held on
benchmark of RL 125·125 m. Rule out the page field book and enter the
above readings. Calculate the reduced level of the points by rise and fall
method and show the usual checks.
- 17.** Write various characteristics of contours.
- 18.** The following details refer to reciprocal levels taken with a dumpy level :

Instrument Station	Staff reading on (m)		Remarks
	<i>A</i>	<i>B</i>	Distance between <i>A</i> and <i>B</i> is 1150 m
<i>A</i>	1·505	2·875	RL of <i>B</i> is 100·00 m
<i>B</i>	0·750	1·895	

Find the following :

- (a) RL of *A*
- (b) Combined error for curvature and refraction
- (c) Collimation error in the instrument

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